

Comprehensive Guide to Cultivating Healthy Corn (Maize): Practices, Growth Stages, and Risk Management

1. Overview

Corn, scientifically classified as *Zea mays* and globally recognized as maize, is a foundational cereal grain that has been cultivated and cherished by humanity for nearly 9,000 years. Originating from wild grasses domesticated by Indigenous populations in the region that is now Mexico, the Taino people originally called it "mahiz," which translates beautifully to "source of life". Today, it has evolved into one of the most critical crops on the planet, serving as a staple food source, livestock feed, and an industrial raw material. Understanding how to cultivate a healthy corn plant requires a comprehensive approach to basic plant biology and soil science, tailored into practices that are accessible and easy to understand for beginners, home gardeners, and seasoned farmers alike.

When we talk about the different types of corn, it is helpful to know that they are grouped by what the inside of the kernel looks like and how it behaves. Most of the corn grown is "field corn" or "dent corn," making up about 99% of the corn grown in the United States. It is called dent corn because as it dries, the starch inside shrinks and creates a small dent at the top of the kernel. This type is mostly used for animal feed or processed into things like syrup and ethanol. On the other hand, "sweet corn" is the type we eat on the cob. It has a natural, wonderful mutation that prevents its sugars from turning into hard starch, keeping it sweet and tender. A healthy corn plant is defined by its ability to progress smoothly and successfully through a highly predictable series of growth and development stages. In the study of plant health, it is essential to distinguish between the concepts of "growth" and "development". Growth refers to the physical increase in the size of the plant—how tall it gets or how big the leaves grow. This is heavily influenced by environmental factors such as soil moisture, nutrient availability, and competition from nearby weeds. Development, conversely, refers to the plant's progression through its life cycle stages, from a germinating seed to a fully mature plant producing grain. Development is primarily driven by the accumulation of heat from the sun, often measured as heat units or Growing Degree Days.

Corn exhibits what experts call "determinate growth". This means that the plant has a strict, predetermined life cycle. It will focus entirely on vegetative growth—producing leaves, stalks, and roots—until it reaches a specific developmental trigger. Once the plant transitions into its reproductive phase to create tassels and ears, it completely stops all vegetative growth. Because of this determinate nature, corn has a highly limited ability to recover from severe stress. If a corn plant experiences a severe lack of water or a lack of nutrients during a critical phase of leaf or ear formation, it cannot simply grow new leaves or add more kernels later in the season. The potential size of the harvest is permanently reduced.

Therefore, maintaining a healthy corn plant is largely an exercise in proactive care and timely intervention. The overarching goal is to provide the plant with optimal conditions—adequate spacing, precise nutrition, sufficient moisture, and protection from pests and diseases—so that it can maximize its natural genetic potential without encountering any roadblocks. This detailed

guide provides an exhaustive, simple-to-understand breakdown of how to visually identify a healthy plant, the risks that threaten it, and the natural, chemical, and everyday cultural remedies required to ensure a successful, bountiful harvest.

2. Symptoms

Monitoring the health of a corn crop requires a thorough understanding of its growth stages. Plant experts divide the life of a corn plant into two main categories: Vegetative (V) stages and Reproductive (R) stages. Observing the specific physical characteristics of the plant at each stage allows you to confirm that the plant is healthy, thriving, and on track for a good harvest.

Indicators of Good Health in Early Vegetative Stages The vegetative stages begin the moment the seed is planted in the dirt and end when the plant produces a tassel at its very top. These stages are numbered according to the number of fully expanded leaves the plant has produced.

- **Emergence (VE):** The very first visible indicator of a healthy corn plant is timely and uniform emergence from the soil. In optimal soil temperatures (above 68°F) and adequate moisture, a healthy sprout will push a protective sheath called the coleoptile through the soil surface in as little as 6 to 8 days. Uniform emergence is a great sign of health; you want all the plants to come up at the same time so they do not compete with each other.
- **First Leaf (V1):** The V1 stage is marked by the complete unfurling of the first true leaf. A key, easy-to-spot indicator of a healthy V1 plant is the shape of this very first leaf: it should have a distinctly oval or rounded tip. This is unique, as all subsequent leaves produced by the plant for the rest of its life will have sharp, pointed tips.
- **Second and Third Leaves (V2 - V3):** By the time the plant has two to three fully expanded leaves (indicated by a visible "collar" or discolored line where the leaf blade wraps around the stalk), significant and healthy changes are occurring underground. The plant is successfully transitioning from relying on the initial roots that came out of the seed to a permanent, strong root system that forms slightly higher up. A healthy plant at this stage will stand perfectly upright and sturdy. It is supported by the strength of tightly wrapped leaf sheaths, even though the actual hard stalk remains entirely below the soil surface at this point.

Indicators of Good Health in Mid-to-Late Vegetative Stages As the plant moves past the early seedling stages, its growth rate accelerates dramatically. This is when the plant builds the machinery it needs to capture sunlight.

- **Sixth Leaf (V6):** The V6 stage is a critical milestone for checking plant health. At this point, the plant's "growing point"—the vital, sensitive tissue from which all new leaves and eventual reproductive organs emerge—moves from its protected space below the ground to above the soil surface. The stalk begins a period of rapid upward stretching. Underground, the permanent root system is now massive, fully functioning, and extremely healthy, making up about one-third of the plant's total weight.
- **Twelfth Leaf (V12):** By the time the plant has twelve leaves, it is generally standing tall, looking thick and robust, and has accumulated roughly 10% of its final dry weight. A key indicator of hidden health at this stage is that the plant is currently determining the size of the ear; a healthy plant with plenty of water and nutrients will prepare a large ear with many potential kernels.
- **Fifteenth to Eighteenth Leaf (V15 - V18):** This is a stage of explosive, rapid growth. The plant may be growing several inches every single day, approaching shoulder height or taller. A visible sign of excellent health and optimal sunlight capture is the architectural

arrangement of the leaves. The uppermost leaves should stand up somewhat vertically, at about a 30-degree angle, while the lower leaves rest at a wider, flatter 45-degree angle. This perfect geometric arrangement allows maximum sunlight to penetrate deep into the crop canopy, fueling the plant's energy factories.

Indicators of Good Health in Reproductive Stages Once the final leaf is produced, the plant enters the reproductive (R) stages, focusing all its gathered energy on creating and filling the kernels that we eventually eat or harvest.

- **Silking (R1):** This is arguably the most critical stage in determining the final harvest. A healthy plant at R1 will have a fully emerged, branching tassel at the very top, shedding millions of tiny pollen grains. Lower down, pale, sticky silks will be emerging from the green husks of the ear. Each individual silk is attached to a single potential kernel inside the husk; pollen from the top must land on the sticky silk and travel down its length to create a kernel.
- **Blister (R2):** Approximately 10 to 14 days after the silks appear, the physical indicators of health change. The previously sticky, pale silks will now turn brown, dry out, and naturally detach from the kernels. If you peel back the husk, the healthy kernels will resemble tiny white blisters filled with a clear fluid.
- **Milk (R3):** About 20 days after silking, the kernels begin to change color. A healthy ear will feature kernels that are distinctly yellow on the outside. If you press your fingernail into a kernel, it will release a milky white fluid, indicating that the plant is successfully and healthily converting its sugars into accumulating starch.
- **Dough (R4):** By roughly four weeks after the silks emerged, the starch inside the kernel continues to gather and thicken. A healthy kernel at this stage will have a pasty, thick, dough-like consistency and will have expanded significantly, turning a darker, richer shade of yellow.
- **Dent (R5):** At the R5 stage, nearly all kernels on a healthy ear of field corn will feature a distinct "dent" at their top. This dent forms because the moisture inside the kernel is decreasing, causing the starchy interior to shrink and pull the top of the kernel inward. A visible "milk line" will also appear on the side of the kernel, acting as a visible boundary between the hard, starchy solid portion at the top and the softer, doughy portion near the cob.
- **Black Layer (R6):** This final stage marks true physiological maturity. The plant has completely finished its life cycle. A healthy maturity is indicated by the formation of a dark, black seal at the very base of the kernel where it attaches to the cob. This black layer acts like a closed door, cutting off all further water and nutrient transfer from the plant to the kernel. The kernels are now at their maximum weight and will naturally dry down, ready to be harvested.

3. Causes & Spread

To maintain the healthy indicators described above, one must understand the myriad of risks that can derail a corn plant's development. Corn health is constantly threatened by a combination of environmental stressors, common farming mistakes, and biological pathogens like fungi and bugs. Plant experts often refer to the "Disease Triangle." This is a simple rule stating that a disease will only break out when three things come together at the exact same time: a vulnerable plant, an active disease-causing germ (pathogen), and the perfect weather conditions for that germ to grow.

Environmental Risks and How to Avoid Them

- **Cold and Wet Soils:** Early in the spring season, planting seeds into soils that are too wet or too cold (below 55°F) poses a severe risk to plant health. Corn seeds require warmth to wake up and sprout quickly. If the seed sits in cold, damp soil, its germination can be delayed for up to a month. This prolonged exposure in the wet dirt leaves the seed highly vulnerable to rotting away before it ever sees the sun. You avoid this simply by waiting for the soil to warm up nicely before planting.
- **Drought and Water Stress:** Because a healthy corn plant grows so incredibly fast, it has an immense thirst for water. The most dangerous time for a corn plant to experience a lack of water is during the period right before and during the silking stage. If the plant is dehydrated when it is trying to push its silks out, the silks will fail to emerge or will dry out too quickly to catch any pollen. This leads to ears with missing kernels or completely bare cobs. Keeping a close eye on rainfall and providing water during this specific window is crucial to avoid losing up to half of your harvest.
- **Nutrient Starvation:** The period from the time the plant is knee-high to the time it produces a tassel is a time of extreme, fast growth. If the soil lacks adequate food during this window, the plant will stunt its growth and turn pale. Because corn cannot pause its life cycle to wait for you to add fertilizer, a lack of food during these stages permanently limits the number of kernels the ear can produce.

Common Pathogens and Diseases Fungi, bacteria, and viruses can spread rapidly through a corn patch under the right weather conditions. Many of these tiny germs are perfectly adapted to outdoor garden and farm environments and can "overwinter"—meaning they survive the freezing winter months by hiding perfectly safe inside the dead corn stalks and leaves left on the soil surface from the previous year.

- **Seed Rots and Seedling Blights:** These problems are caused by various common soil-dwelling fungi with complicated names like *Fusarium* and *Pythium*. These germs attack the germinating seed and the fragile young root systems. They love the cool, wet soil conditions mentioned earlier. Infected baby plants may fail to emerge at all, or they might come up looking yellow, stunted, and weak enough to fall over.
- **Foliar Diseases (Leaf Blights and Rusts):** Fungal infections that attack the leaves are highly destructive because the leaves are the plant's solar panels. If the leaves are damaged, the plant cannot create energy.
 - **Southern Rust:** This fungus requires warm, wet, and highly humid weather to spread. It shows up as thousands of tiny, orange or rust-colored blister-like bumps on the surface of the green leaves. These bumps are actually tiny explosions of fungal dust ripping through the plant tissue. By tearing thousands of tiny holes in the leaves, the fungus severely damages the plant's health, causing the leaves to die early and resulting in very small ears.
 - **Gray Leaf Spot and Anthracnose:** These are other common fungal sicknesses that cause distinct brown or gray dead spots on the leaves. These spots usually start on the lower leaves and progressively move up the plant, robbing the plant of its green, energy-producing areas. They spread when raindrops splash the fungal spores from the ground up onto the lower leaves.
- **Stalk and Ear Rots:** Toward the end of the season as harvest approaches, certain germs can invade the main stalk, rotting it from the inside out and causing the plant to snap and fall over before you can pick the corn. Ear rots infect the kernels directly on the cob, turning them moldy, ruining the quality, and sometimes making them unsafe to eat.

Human Health and Dietary Risks While corn is generally healthy, there are a few risks to be aware of regarding human consumption. For people who have Celiac disease—an autoimmune

disorder where eating gluten causes severe bodily reactions—corn is generally considered safe, but some individuals find that corn proteins can still trigger uncomfortable issues for them. Additionally, eating large amounts of corn can cause flare-ups in symptoms for people who suffer from Irritable Bowel Syndrome (IBS) due to how it digests. Regarding Genetically Modified (GM) corn, which scientists alter slightly to resist bugs or survive droughts, some consumers worry about safety. However, current evidence shows no risk to human health from eating these varieties, though many still prefer organically grown options.

4. Treatment / Remedies

Ensuring the health of a corn plant requires a proactive approach to feeding the plant and managing pests. Depending on your personal gardening philosophy or farming goals, these treatments can be entirely natural and organic, using materials found in nature, or they can involve the precise, careful application of chemical plant foods and protective sprays.

Natural / Home Remedies Organic corn farming and gardening rely on a system that mimics natural ecological processes to build up the soil's health and manage pests without using any synthetic or artificial chemicals. The main idea here is simple: if you feed the soil and make it healthy, the soil will naturally feed and protect the plant.

- **Building Soil with Compost and Manure:** Animal manures are an excellent, traditional, and natural remedy for poor soil. They provide a rich source of nitrogen, phosphorus, and essential organic matter that makes the soil dark and crumbly. In traditional farming systems in places like Nepal, applying about 50 to 60 baskets of well-fermented farmyard manure for a specific area of land provides a wonderful, slow-release food base for the plants. To get the absolute most benefit and prevent the valuable nutrients in the manure from evaporating into the air, you should use a trick called "sheet composting". This simply involves spreading the manure and immediately mixing it or plowing it into the dirt. Mixing it in right away saves a huge amount of nutrients from being lost to the wind.
- **Using Rock Minerals:** To correct soil that is unbalanced naturally, you can apply simple, crushed rock powders. Agricultural lime is a rock powder used to sweeten soils that have become too sour or acidic, while also adding calcium. Soft-rock phosphate provides a natural source of phosphorus for strong roots, and gypsum is a great mineral to sprinkle on hard, dense clay soils to help break them up and make them fluffy.
- **Natural Pest Sprays (Neem Oil):** When pesky insects threaten to eat your crop, natural remedies can be highly effective. A very common and popular botanical spray is Neem oil. This oil is pressed from the seeds of the Neem tree, which is native to Asia. It works like magic in an organic garden because it contains a natural compound that makes insects stop eating immediately. When bugs take a bite of a leaf sprayed with Neem oil, their stomachs are upset, and they lose their appetite and ability to grow, saving the plant without harsh chemicals.
- **Friendly Bacteria (Bt):** *Bacillus thuringiensis* (Bt for short) is a naturally occurring bacteria found in dirt that acts as a safe, natural bug killer. You can buy it as a safe dust. When you sprinkle this dust into the top funnel of the corn plant where caterpillars like to hide, it waits for them. Once a caterpillar eats a tiny bit of the dust, the bacteria give the caterpillar a fatal stomach ache, protecting your corn.

Chemical Remedies For those using conventional methods, maintaining health involves the precise, calculated application of commercial fertilizers to match the plant's exact needs, as well as the use of safe synthetic agents to protect the young plants from weeds.

- **Calculated Plant Food (Fertilizer):** High-yielding modern corn needs a lot of food,

especially nitrogen. Experts use a simple math rule: the plant needs about 1.37 pounds of nitrogen for every bushel of corn grain you hope to harvest.

- **Starter Fertilizers:** To give the young baby plant a safe, immediate boost, growers use a "starter fertilizer" at planting time. Because putting strong chemical fertilizer directly on a seed will burn and kill it, the fertilizer must be placed carefully. The best method is to place the fertilizer in a band in the dirt exactly two inches to the side of the seed and two inches below it. This is called a "2x2 placement". This ensures the young roots reach the food just as they need it, without risking chemical burn.
- **Split Feeding (Side-Dressing):** Applying all the fertilizer at planting is a bad idea; heavy spring rain can easily wash the food deep into the soil and away from the roots before the plant can eat it. Instead, healthy management requires giving the plant its food in smaller meals. After the starter fertilizer, a second dose is applied when the corn is about knee-high (12 to 15 inches tall). In highly advanced farm systems, liquid fertilizer is simply injected directly into the irrigation water in small, weekly doses, spoon-feeding the plant right up until it produces a tassel.
- **Safe Weed Control:** Corn hates weeds because weeds steal its water and food. Safe herbicides, like carefully measured mixtures of atrazine, can be sprayed on the bare dirt within three days after planting the seeds. This creates a clean, weed-free zone for the baby corn to emerge into without having to fight for sunlight. Secondary nutrients like sulfur (20-30 pounds per acre), boron, and zinc can also be applied based on what a soil test tells you is missing.

5. Preventive Measures

Treating a problem after it appears is always much harder than preventing it from happening in the first place. Good gardening and farming practices can create an environment where corn plants naturally thrive and sickness struggles to survive.

Crop Rotation and Field Cleaning The single most powerful, natural way to prevent corn diseases and pests from taking over your garden or farm is to practice crop rotation. This means you never plant corn in the exact same spot two years in a row. By switching corn out for a completely different type of plant—like beans, peas, or alfalfa—you starve the germs and bugs that love corn. Because these pests wake up the next year and find a plant they cannot eat or infect, their populations die off naturally.

Going hand-in-hand with rotation is field cleaning, or sanitation. Many harmful leaf diseases and insect larvae survive the freezing winter by hiding inside the dead, woody stalks of the old corn plants left in the field. By shredding these old stalks with a mower and plowing them deep into the dirt at the end of the year, you destroy their winter homes, drastically reducing the chance of disease returning the following spring.

Proper Spacing and Planting Corn plants are very sensitive to being crowded. If you plant them too close together, they will fight each other for sunlight, water, and food, resulting in very thin, weak plants with tiny ears.

- **Row Spacing:** For optimal health and airflow, corn seeds should be planted in rows that are spaced 30 to 38 inches apart (about 75 centimeters).
- **Plant Spacing:** Within that row, each individual plant should be spaced 8 to 12 inches apart, making sure to drop exactly one seed into each hole.
- **Planting Depth:** How deep you bury the seed is crucial for the roots to grow properly. If you have heavy, sticky clay soil that holds water, plant the seeds shallow, about 1 inch deep. If you have nice, normal garden soil, 1.5 inches is perfect. If your soil is very sandy

and dries out fast, plant the seeds 2 inches deep so they stay surrounded by wet dirt.

Always making sure the seed is at the right depth ensures the roots grab the earth tightly.

Companion Planting For home gardeners and organic growers, companion planting is a brilliant and beautiful way to protect corn. This means planting different species right next to each other because they help each other grow.

- **The Three Sisters:** This is an ancient Indigenous method of planting beans, squash, and corn together. The beans are amazing because they have a special ability to pull nitrogen food out of the air and put it into the soil, acting as a free fertilizer for the hungry corn. Meanwhile, planting sprawling squash vines at the base of the corn acts like a living blanket over the dirt, shading the soil to lock in water and smothering any bad weeds trying to grow.
- **Protective Flowers and Herbs:** Planting Marigolds around your corn acts as an invisible shield in the dirt. Marigold roots release special juices that chase away harmful microscopic worms called nematodes that like to chew on corn roots. Herbs with strong smells, like Basil, onions, and garlic, confuse and chase away flying bugs and aphids. Planting bright sunflowers on the edges of the corn patch is also a great idea; they act as a sturdy fence against strong winds, protecting the delicate corn stalks from snapping, while also attracting bees and good bugs that eat the bad bugs.

Choosing Resistant Seeds Before you even touch a shovel, you can prevent disease by buying the right seeds. Plant breeders have developed special corn seeds that have a natural toughness or immunity against specific local diseases. You should ask your local seed seller for varieties that are known to be strong against the diseases common in your specific area. Be careful not to just buy the seed that promises the absolute biggest harvest, because sometimes those "racehorse" seeds put all their energy into making corn and forget to build up their immune system against sickness.

6. Suggestions & Best Practices

Beyond the basic rules of planting and watering, the difference between an okay garden and an exceptional, high-yielding harvest comes down to everyday management during the summer. Using these practical tips ensures the plant stays healthy from the day it sprouts to the day you pick it.

Smart Watering and Irrigation A healthy, tall corn plant is incredibly thirsty. To maintain its perfect health, you must make sure it gets enough water, especially during the crucial weeks when it is making the ear.

- **When to Water:** You should not just water randomly. It must be based on how big the plant is and how dry the dirt feels.
- **Early Season:** If the top of the dirt is dry right after you plant the seeds, give it a light sprinkle of water (about half an inch) to help the seeds wake up evenly.
- **The Most Important Time:** The absolute most critical time to water your corn is when it is growing its tassel and pushing out its silks. If you let the corn dry out during this week, you will lose a huge amount of your harvest. If it has not rained, you must soak the ground deeply to prevent the silks from drying out and the kernels from dying.

The Art of Crop Scouting You cannot fix a problem if you do not know it exists. "Scouting" is the simple practice of regularly walking through your garden or field to check on the plants, look for bugs, and see if the leaves look healthy.

- **When to Look:** You should walk through your corn a few times a season: right after the plants pop out of the ground to make sure they look okay; every two weeks during the

summer to check for bug holes or strange spots on the leaves; and right before harvest to make sure the stalks are strong.

- **How to Walk the Field:** Do not just look at the plants on the very edge of the garden. You need to walk into the middle. Experts recommend walking in a zig-zag pattern or a big "W" shape through the patch. Stop a few times along the way and look very closely at the leaves of 50 to 100 different plants.
- **What to Look For:** Look for patterns. If all the plants in the entire garden are turning slightly yellow, they probably just need some nitrogen fertilizer. But if the plants are only yellow and sick in one low, muddy corner of the garden, the problem is likely just too much water, not a lack of food. Identifying these patterns helps you make smart choices. You can even use free phone apps like "Soil Web" to help you understand your dirt better.

Soil Testing Mid-Season Guessing how much plant food your corn needs is a bad idea.

- **Baseline Testing:** Before you plant in the spring, take a small cup of dirt to a local garden center or university extension for a soil test. This will tell you your starting pH level (which should ideally be around 6.5 for corn) and what minerals are missing.
- **The Knee-High Test:** When the corn reaches the V6 stage (about knee-high), a great trick is to take another small soil sample to measure exactly how much nitrogen is currently in the dirt. This test takes all the guesswork out of your mid-summer feeding, telling you exactly how much extra fertilizer you need to add to finish the crop perfectly, which saves you money and keeps the plants perfectly healthy.

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8. Key Takeaways

- **Understand How Corn Grows:** Corn has a strict timeline and cannot grow new leaves or fix its ears once it starts reproducing. Taking great care of the plant while it is young and during the time it makes its silks is the absolute most important rule for a great harvest.
- **Look for Healthy Signs:** You can tell your corn is doing well if it all sprouts at the same time, if the stalks start stretching up quickly when they are knee-high, and if the leaves angle up nicely to catch the sun. A healthy ear will show yellow kernels that get thick and doughy, eventually forming a tiny dent at the top and a black seal at the bottom when they are fully ripe.
- **Stop Diseases Before They Start:** Fungi and germs love to hide in the dirt and old, dead leaves. Prevent sickness by never planting corn in the exact same spot two years in a row, choosing strong seeds, and cleaning up your garden at the end of the year.
- **Feed and Space Plants Properly:** Corn is a very hungry plant. Give it nitrogen food in small amounts over the summer rather than all at once so it doesn't wash away. Make sure your rows are spaced about 30 inches apart so the plants don't fight each other for sunlight.
- **Use Nature to Protect the Crop:** Try companion planting by growing beans to add free fertilizer to the dirt, squash to shade the ground, and flowers like marigolds to chase away bad bugs and worms naturally.
- **Water at the Right Time:** Corn drinks a massive amount of water. The most critical time you must not let the dirt dry out is right when the plant is pushing out its tassels and silks; otherwise, you will end up with ears missing most of their kernels.
- **Walk Your Garden (Scouting):** Make a habit of walking a zig-zag path through your corn every couple of weeks. Look closely at the leaves for bug holes or weird spots so you can fix problems early before they ruin your whole crop.

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